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***Kysor of Cadillac, Inc. Site
Wexford County, Michigan
First Five Year Review***

I. Introduction

EPA Region 5 has conducted a five year **review** of the remedial actions implemented at the Kysor of Cadillac, Inc. Site in Wexford County, Michigan. This five year review represents the first review for the Kysor Site. This review was conducted from May 15, 2000 to July 19, 2000 and a site visit was conducted on July 19, 2000. This site visit was attended by Dion Novak, Remedial Project Manager, Larry Campbell, City of Cadillac Utilities Department, Pete Daukss, Tetra Tech-city of Cadillac contractor, and Jim Skipper, Michigan Department of Environmental Quality. This report documents the results of this review. The purpose of five year reviews is to determine whether the remedy at the site **remains** protective of human health and the environment. The methods, findings, and **conclusions** of these reviews are documented in five year review reports. In addition, five year **review reports** identify deficiencies found during the review, if any, and identify **recommendations** to address them.

This review is required by statute and is a **Level 1** review. EPA must implement five year reviews consistent with the Comprehensive Environmental Response, Compensation and Liability Act (CERCLA) and the National Oil and Hazardous Substances Pollution Contingency Plan (NCP). CERCLA Section 121 (c) as amended states:

If the President selects a remedial action that results in any hazardous substances, pollutants or contaminants **remaining at the site**, the President shall review such remedial action no less often than each **five years** after initiation of such remedial actions to assure that human health and the **environment** are being protected by the remedial action being implemented.

The NCP Part 300.430 (f)(ii) of the Code of Federal Regulations (CFR) states:

If a remedial action is selected **that results in** hazardous substances, pollutants or contaminants remaining at the **site above** levels that allow for unlimited use and unrestricted exposure, the lead **agency** shall review such action no less often than every five years after the initiation of **the remedial action**.

This is the first five year review conducted **for the Kysor Of Cadillac, Inc. Site**. The triggering action for this statutory review is the **date of actual** on-site remedial action construction, which was September 1995. Because hazardous **substances** remain at the site above levels that allow for unlimited use and unrestricted exposure, a five year review is required by statute.

II Site Chronology

Table 1 below lists the chronology of events for the Kysor Site.

Table 1

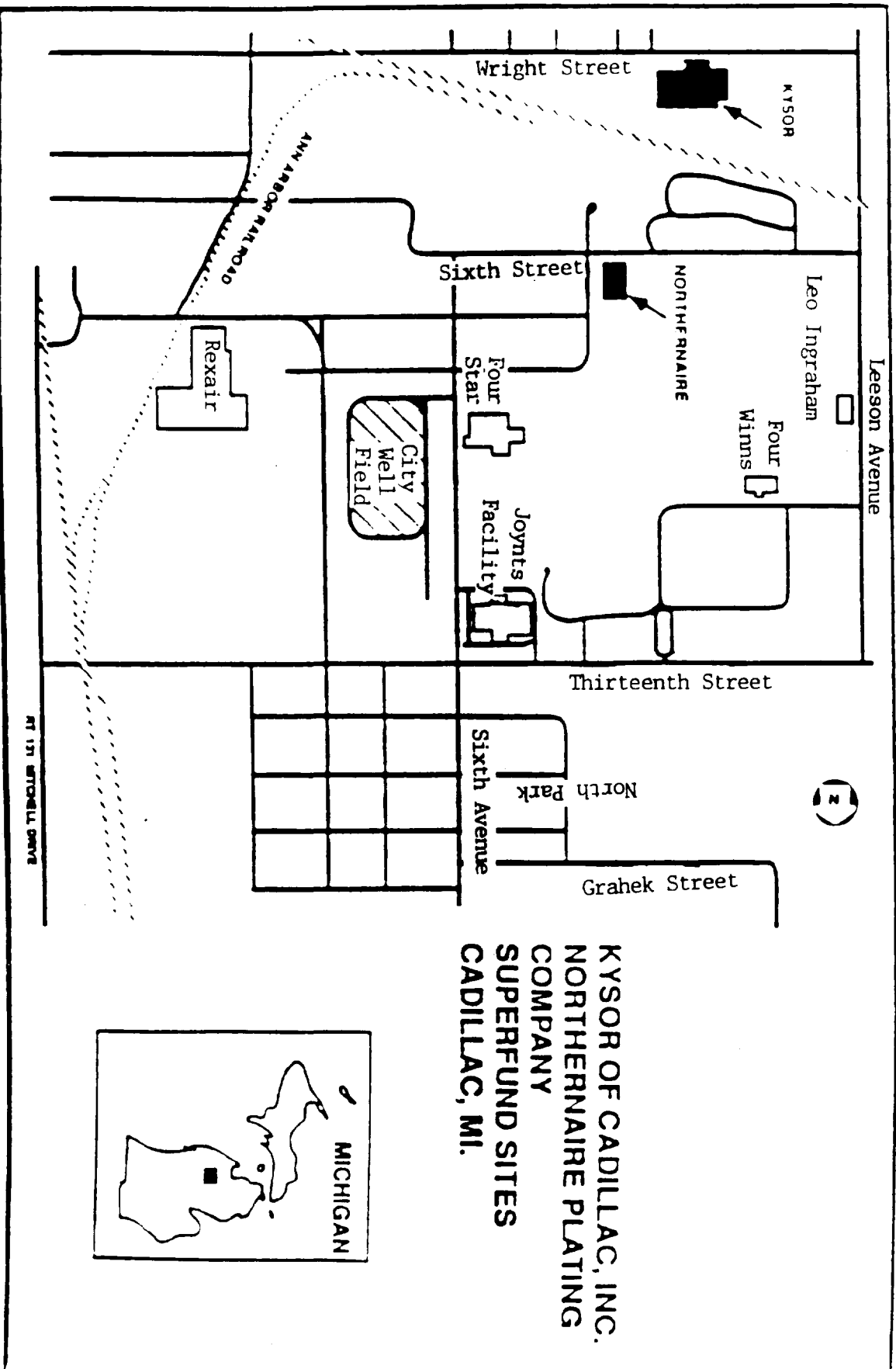
<u>Date</u>	<u>Event</u>
1981	Partial remediation performed by Kysor
9/85	Kysor Site proposed for National Priorities List (NPL)
6/89	Michigan Department of Natural Resources (MDNR) contractor completes Cadillac Area Groundwater Contamination Feasibility Study
9/29/89	Record of Decision (ROD) signed for Kysor/Northernmaire commingled groundwater contamination and Kysor soil contamination
10/4/89	Kysor Site finalized on National Priorities List
5/16/90	Unilateral Administrative Order (UAO) issued for remedial design
3/3/94	Explanation of Significant Differences (ESD) signed increasing area of groundwater cleanup
1/30/95	UAO issued for remedial action
3/6/95	Final design approved

III Background

The Kysor Industrial Corporation Site (Kysor) covers approximately one square mile of the Cadillac Industrial Park within the city of Cadillac, Michigan and an adjacent area north of 13th Street in Haring Township. Kysor has an active facility at 1100 Wright Street at the southern edge of the industrial park, which manufactures truck parts and uses a variety of solvents, degreasers, and paints. It began operations in approximately 1959. Past disposal practices included dumping barrels of spent solvents directly on the soils behind the plant. Partial remediation of the contaminated soils at the Wright Street facility was conducted by Kysor in 1981, including the removal of the top six feet of an area of soil, for a total removal of 700 cubic yards of soil. However, MDNR observers noted strong odors and more than 100 parts per million of several volatile organic compounds (VOCs) remaining in soils. The excavation was backfilled with clean material (See Figure 1).

The site was proposed for the NPL in September 1985 and listed in 1989. The Remedial Investigation/Feasibility Study was conducted by contractors for the State of Michigan. Soils adjacent to the plant remained contaminated with a variety of VOCs, including trichloroethylene (TCE) up to 74 ppm (cleanup level 0.07 ppm), trichloroethane (TCA) up to 24 ppm (cleanup

Figure 1



level 7.6 ppm), and xylenes up to 520 ppm (cleanup level 141 ppm). Although the risk from the contaminated soils were not assessed quantitatively, groundwater monitoring data show them to be a continuing source of groundwater contamination. Groundwater downgradient of the Kysor Site also contained elevated levels of VOCs, including TCE up to 78 ppm (drinking water limit of 0.005 ppm), TCA up to 32 ppm (drinking water limit of 0.2 ppm), and tetrachloroethylene (PCE) up to 0.72 ppm (drinking water limit of 0.001 ppm). These contaminants led to a cumulative excess lifetime cancer risk up to 5×10^{-2} .

The groundwater contamination from the Kysor Site is co-mingled with that from other facilities downgradient of it in the Cadillac industrial park, including the Northern Plating Superfund Site. VOCs from the Kysor Site and other facilities have spread northward throughout the industrial park and vertically into the shallow and intermediate aquifers at the Site. Chromium contaminated groundwater from the Northern Plating site has also spread northward and vertically into both aquifers, co-mingling with VOC contaminants from the Kysor Site. One plume cannot be extracted and treated without impacting the other.

A Local Development Financial Authority (LDFA) was created by the City of Cadillac to finance the construction of the remediation facilities in accordance with the remedial design. The City of Cadillac Utilities Department performs the operation and maintenance services associated with the RA for soil and groundwater for the Site.

IV Remedial Actions

Remedy Selection/Objectives

9/29/95 - 8/21/96

Construction of site remedy which includes an 18 well extraction system with collection and discharge piping, a groundwater treatment system utilizing carbon adsorption to treat hexavalent chromium and air stripping to treat VOCs with surface water discharge to the Clam River, a soil vapor extraction system with vapor phase carbon adsorption to treat soils contaminated with VOCs, and removal of approximately 1000 yards of contaminated soil not in the September 1989 ROD but required by the January 1995 UAO.

Remedy Implementation

The design for the site remedial actions was completed in March, 1995. Construction of the site remedial actions began in September, 1995 and was completed in August, 1996. The design and construction of the remedial actions were performed under the terms of two UAOs.

Static water level data since 1996 have indicated that the groundwater plume has been adequately contained with the 18 well extraction system. This is based on water level monitoring, which

occurred on a quarterly basis for the **first year of operation** and on an annual basis since late 1997. Contaminant concentration data for VOCs and hexavalent chromium have consistently decreased since startup of the extraction system. In July 1997, the granular activated carbon contactor was taken off-line because hexavalent chromium had broken through the contactors. This problem was corrected and the contactor was placed back in service in December 1997. Since that time, the effluent from the treatment system has not exceeded Clam River discharge standards (See Figure 2).

System Operations/O&M

The groundwater extraction system has operated continuously since startup with no problems encountered. All of the extraction wells have operated continuously since being placed on-line. Weekly samples are collected of extraction system influent and effluent and analyzed for verification of compliance with system discharge limits.

Progress Since the Last Five Year Review

This is the first five year review for the Site.

V Five Year Review Findings

The groundwater cleanup is in place and has been operational since August 1996. The remedial goals for the Kysor cleanup were to construct a groundwater extraction and treatment system to remediate contaminated groundwater at the site (See Figure 1). Groundwater treatment includes carbon adsorption for hexavalent chromium treatment and packed tower air stripping for VOC contamination. Soil treatment for VOCs consists of a soil vapor extraction system consisting of 23 extraction/induction wells that direct soil vapors to a vapor phase carbon treatment system.

Figure 1 Groundwater cleanup standards

<u>Compound</u>	<u>Cleanup Standard</u>
1,1,1-TCA	200 ppb
trans 1,2-DCA	70 ppb
1,1-DCE	5 ppb
1,2-DCA	5 ppb
methylene chloride	5 ppb
PCE	1 ppb
TCE	5 ppb
xylenes	440 ppb
toluene	40 ppb

Figure 2 Effluent criteria for discharge to the Clam River

<u>Compound</u>	<u>Discharge Criteria</u>
1,2 - dichloroethane	5 ppb
1,1,1-trichloroethane	5 ppb
trichloroethene	5 ppb
tetrachloroethane	5 ppb
hexavalent chromium	8.3 ppb
total chromium	59 ppb

Figure 3 Soil cleanup criteria

<u>Compound</u>	<u>cleanup criteria</u>
trichloroethylene	0.07 ppb
xylene	141 ppb
1,1,1 trichloroethane	7.6 ppb
toluene	724 ppb

Soil cleanup levels were calculated so that residual contamination in the soil would not result in a continuing source of contamination to the groundwater above groundwater cleanup standards (See Figure 3).

The soil vapor extraction system is designed so that the vapors are extracted from four wells at any one time with the other 19 wells being used as air induction wells to accelerate VOC removal from the soils. The system was initially designed with two vapor phase carbon contactors but in July 1997, an additional two contactors were brought on-line to extend the period of time between carbon changeouts. One of these was taken off-line in August 1997, leaving three remaining. Analysis of the spent carbon has indicated that it is not a characteristic hazardous waste.

The annual report for 1997 recommended that, because the extraction system was achieving the performance objective of groundwater plume containment, the groundwater monitoring frequency be changed from quarterly to annually, which was determined to be sufficient to monitor the status of groundwater in the Site area. This annual monitoring began with the October 1998 sampling event for the compliance monitoring wells. A select group of monitoring wells designated by the EPA are still monitored quarterly.

A request was received from the City of Cadillac that requested a modification to the September 1989 Record of Decision for the cleanup standard for tetrachloroethene (PCE) in February 1999. The City of Cadillac requested that the cleanup standard for PCE for the site be changed from 1 ppb to 5 ppb, to be consistent with the standard that the Michigan Department of Environmental Quality (MDEQ) was using on another cleanup site.

VI Assessment

The groundwater extraction system continues to provide containment of the contaminated groundwater plume, as evidenced by **water level data** that is continuing to be collected. This containment continues to be documented by **groundwater data** presented in annual performance monitoring reports. As mentioned above, **VOC concentrations** have generally declined in the shallow aquifer and have remained relatively constant in the intermediate aquifer. Hexavalent chromium concentrations have declined to **below** cleanup criteria in the latest sampling event (See Figure 1).

Groundwater treatment has continued to **reduce** the amount of VOC and hexavalent chromium contamination in the underlying aquifer. **Influent** concentrations to the treatment facility have decreased as a result of operations indicating that the system is meeting design standards. The treatment system has consistently **achieved compliance** with NPDES effluent limits (See Figure 2) for discharge to the Clam River.

The soil vapor extraction system continues to effect treatment of contaminated soil to achieve ROD cleanup standards (See Figure 3). **During** the last year of operation, approximately 152 pounds of VOCs as methane have been **removed** from the soil remediation portion of the Site.

VII Deficiencies

At present, the site remedy is performing **adequately** and achieving ROD performance standards. There are no deficiencies at present with **any component** of the implemented remedy.

VIII Recommendations and Required Actions

This five year review report has **summarized** the current remedial activities at the Site and also describes the future monitoring activities at the Site. The following actions are recommended for continued operation and maintenance of **site remedies**.

The groundwater extraction and treatment **system** should continue to be operated, providing downgradient containment of the **VOC and hexavalent chromium** contaminant plumes. It is recommended that the current **groundwater monitoring** data be studied to prepare a trend analysis report to evaluate contaminant trends in **groundwater** near the Site. This report should also include an analysis of the current **monitoring network** to determine the optimal configuration and frequency for Site remedy monitoring. **The results** of this analysis can be used to evaluate the performance of the extraction system at **reducing** contaminant levels and also be used to evaluate potential contaminant contributions from **neighboring sites**.

It is also recommended that the **groundwater** extraction and treatment system be studied to

determine if it can undergo optimization to improve performance while maintaining its protectiveness.

The results of the soil vapor extraction system should be evaluated to determine the effectiveness of soil remediation and to determine whether the current system configuration can be optimized. Soil confirmation sampling to support soil vapor extraction system shutdown should also be proposed to the Agencies in this report.

It is recommended that close coordination be established with the ongoing studies of neighboring sources of potential groundwater contamination, such as the Mitchell site, the Rexair site and the Paulstra CRC Sites with the Michigan Department of Environmental Quality so that any adverse impacts on the Kysor/Northernair system can be quantified and prevented.

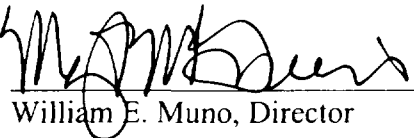
It is also recommended that the February 1999 proposed change in the cleanup standard for PCE be assessed and resolved between the EPA and the MDEQ, with results communicated to the City of Cadillac.

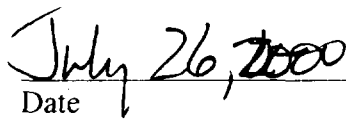
IX Protectiveness of Remedy

The remedy for the Kysor Site is protective of human health and the environment.

X Next Review

The next five year review will be conducted within five years of the completion of this report, which will be July, 2005.

for 
William E. Muno, Director
Superfund Division
U.S. EPA Region 5


Date